ALICE L WALTON SCHOOL OF MEDICINE

Pharmacokinetics1 – 2 of 3 Drug Metabolism and Excretion

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Adapted from: Drug Administration, Metabolism, and Excretion

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Distinguish between Phase I and Phase II reactions in hepatic drug metabolism.
2. Determine which drugs act as substrates and identify the key enzymes implicated in their hepatic biotransformation.
3. Explain the r
4. Identify relevant pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.

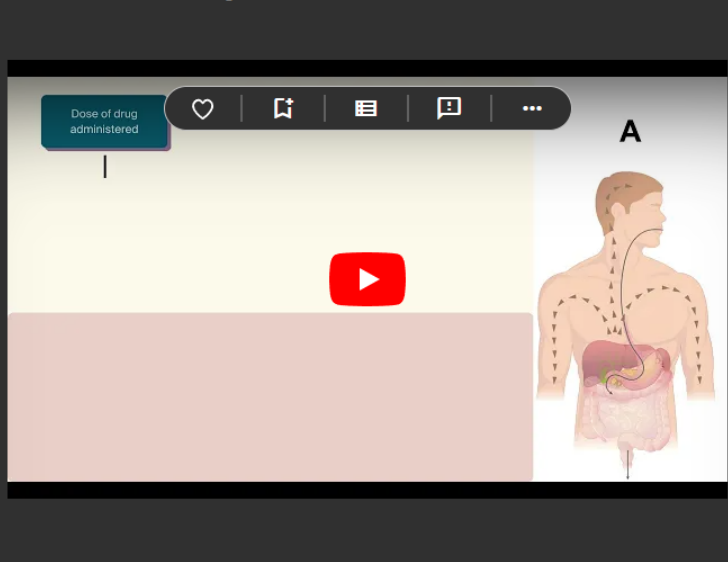
CASE CONNECTION

You are a first-year medical student on rotation in an outpatient clinic. AK is a 40-year-old male who is visiting the clinic. AK has been recently diagnosed with hypertension and high cholesterol (hypercholesteremia). The primary care physician at the clinic is making recommendations for drugs to AK. He wants AK to start taking verapamil (A calcium channel blocker) and Simvastatin (A statin for high cholesterol). He counsels AK on the use of the new medications and strictly asks him not to take grapefruit juice. AK, likes drinking juice and is not pleased. He says, "Grapefruit juice, what... it's harmless.. it's a juice?"

How will you explain to AK the use of grapefruit juice with the other medications? Consider your answer as you read, and we'll revisit AK at the end of the brick.

GO TO CONCLUSION ↓

Recall that drug administration, metabolism, and excretion (ADME) are what happens to a drug when it enters the body, including what the body does to it (Figure 1). This is also known as **pharmacokinetics**. This brick focuses on the metabolism and excretion processes.



Review: Which type of drug administration yields the highest bioavailability?

What is Drug Metabolism? (LO#1)

Once a drug is in the body, a number of modifications to it are possible. Sometimes the drug circulates, either on its own or bound to a protein. However, the body immediately starts working on many drugs, sometimes making them more active and sometimes less. The most common modification is making a drug easier to excrete.

One major role for drug metabolism is to transform drugs into compounds that can be more easily eliminated from the body in the urine or feces. The main organ in the body that metabolizes drugs is the liver, which contains a high concentration of drug-metabolizing enzymes. However, every biologic tissue has some drug-metabolizing capability, including the GI tract, kidneys, lungs, and skin.

The goal of this type of drug metabolism is to render the drug inactive and water soluble. Why? Making the drug inactive prevents prolonged drug effects, and water solubility allows for urinary excretion through the kidneys, the most common route of drug excretion. From the pH/pKa brick, we know that polar drugs or metabolites that are hydrophilic (or like water) do not cross the lipid barriers; hence are easier to excrete. The products of drug metabolism, in broad terms is to make the drug polar, so that they can be excreted.

What is the primary goal of drug metabolism?

Based on the route of administration, when drug is absorbed, it may or may not pass through an extensive first-pass metabolism. Everything that enters the bloodstream, whether swallowed, injected, inhaled, absorbed through the skin, or produced by the body itself, is metabolized by the liver. These chemical reactions that alter the drug are known as biotransformation. Liver biotransformation is performed by liver enzymes and are categorized into either **Phase I** (modification) and **Phase II** (conjugation).

What are two effects of drug metabolism?

What are the phases of hepatic biotransformations? (LO#1and2)

Let's now take a closer look at the two basic phases of drug metabolism, based on their type of enzymatic reaction.

Phase I Biotransformation (modifications): These type of biotransformations (modifications) occur by the biochemical reactions of reduction, oxidation, or hydrolysis. Phase I drug reactions are carried out by liver enzymes. These enzymes are called **cytochrome P450 (CYP450) (or simply CYPs)** proteins. CYP450 proteins are a family of enzymes that work by oxidizing lipophilic (fat-loving) compounds to alter or terminate their activity. The CYPs are located in the smooth endoplasmic reticulum of the hepatocytes. The term "P450" refers to the spectrophotometric wavelength, 450 nm, at which these enzymes are at their absorption maximum. These CYPs alter the chemical structure of the drug hence the terms **'modifications' or 'biotransformation'** are associated with Phase I reactions. The modified chemical form of the drugs are called **metabolites (or drug metabolite).** (Figure 2)

Drug metabolites in most cases are less active than the parent compound, hence the liver is known as the "detoxifying" organ. CYP proteins also play an important role in catalyzing reactions involved in synthesizing endogenous substances like cholesterol, bile acids, vitamin D, and steroid hormones. All CYP450 proteins utilize heme as a cofactor, which is why they primarily transfer electrons (oxidation).

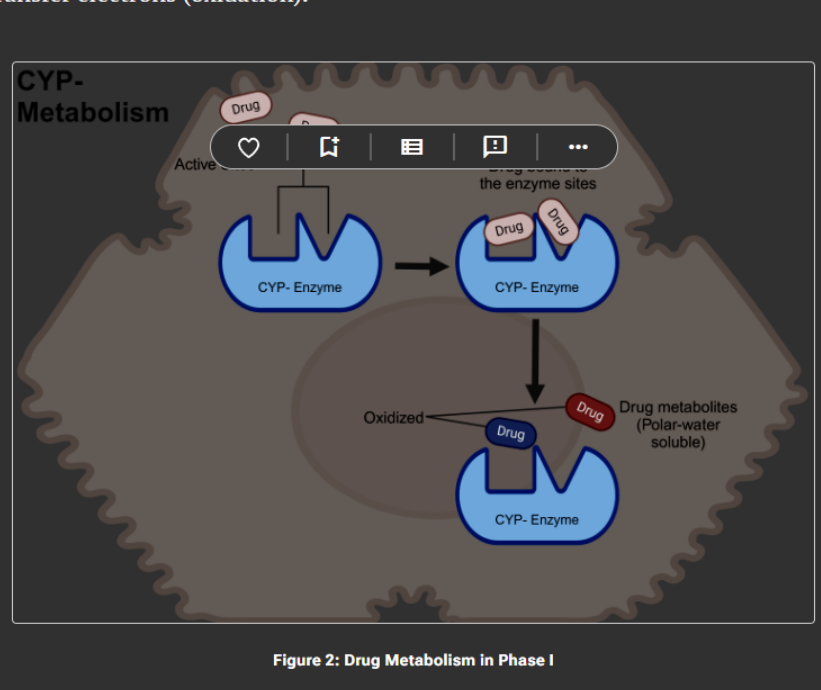
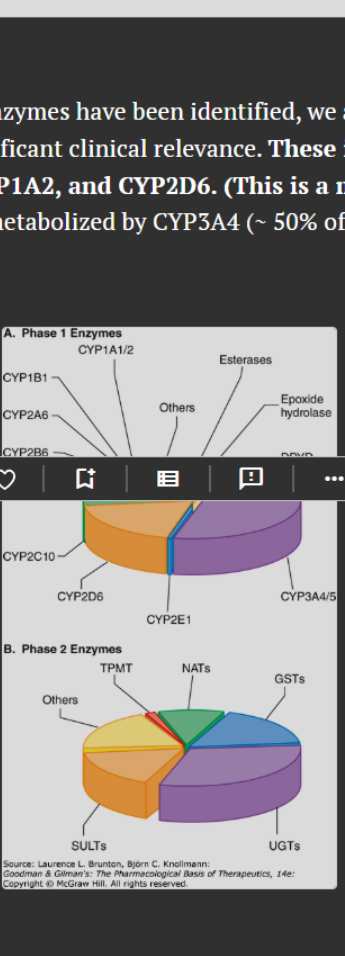


Figure 2: Drug Metabolism in Phase I

Mnemonic

Red-Ox

While numerous CYP enzymes have been identified, we are interested in the isoforms that have significant clinical relevance. **These include: CYP3A4, CYP2C9, CYP2C19, CYP1A2, and CYP2D6. (This is a must know list!!).** Most drugs are metabolized by CYP3A4 (~ 50% of the prescription drugs).



Prodrugs

Although most drugs are inactivated by hepatic drug metabolism, conversely, a few drugs rely on liver metabolism to be converted into their physiologically active form. These agents are called **prodrugs**. A **prodrug** is a biologically inactive compound that undergoes metabolism to be activated. One common example is the acid reflux drug, omeprazole. Omeprazole is a prodrug, that is activated after it goes chemical reactions. On the other hand, drugs such as codeine are modified by CYP enzyme to its active form (morphine).

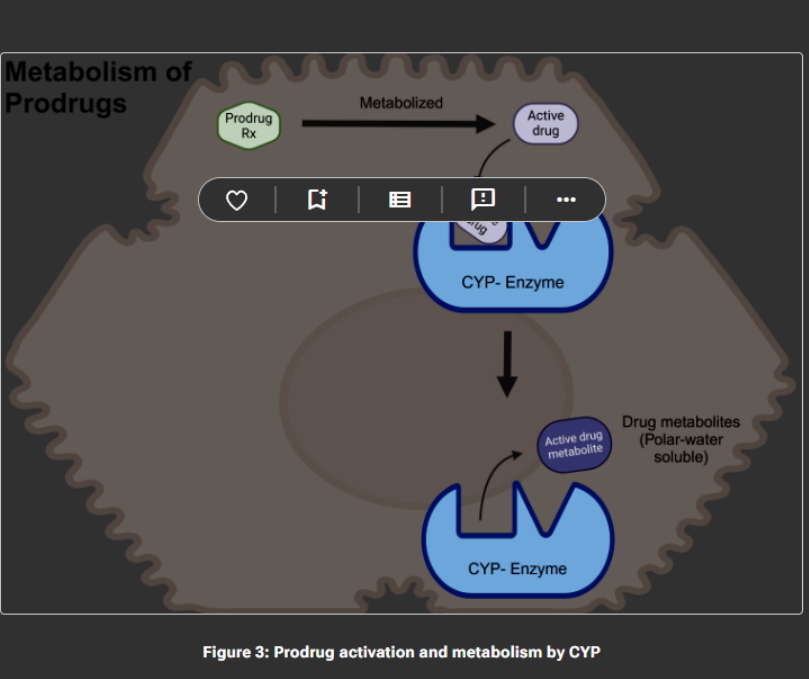


Figure 3: Prodrug activation and metabolism by CYP

How do CYP proteins chemically alter drugs?

Phase II Conjugation. This type of biotransformation involves enzymes adding a functional chemical group to the drug in a process called conjugation. This type of reaction is carried out most commonly by hepatic transferase enzymes. It results in two types of pharmacologically inactive metabolites: polar molecules (water soluble) that can be excreted in the urine and nonpolar molecules that are eliminated in the feces. Phase II reactions can occur in the liver, kidneys, lungs, intestines, and other organs. Drugs can enter phase II reactions directly or after phase I biotransformation. *Sometimes, phase I reactions must happen to a drug before phase II reactions can occur.*

The adjuncts that are conjugated are amino acids, glucuronic acid, glutathione, acetate, or sulfate. These adjuncts are added via methylation, glucuronidation, acetylation, or sulfation reactions. The most common pathway for Phase II conjugation is glucuronidation.

What is the difference between phase I and phase II biotransformation?

CLINICAL CORRELATION

Hepatic diseases, like liver cirrhosis, impact the elimination of drugs processed by the liver. When phase I and II metabolic pathways are compromised, the half-lives of long-acting drugs can increase, raising the risk of toxicity. Additionally, a decline in plasma protein levels due to liver disease can result in more unbound drugs in the plasma. As liver disease worsens, drug dosages must be reduced to achieve the same therapeutic effect. Furthermore, changes in liver perfusion can influence drug metabolism. Conditions that decrease blood flow to the liver, such as shock, hypovolemia, or hypotension, can reduce the metabolic rate.

What Can Affect Drug Metabolism? (LO#2 and 4)

While many factors can affect drug metabolism in phase I, the two main factors are, 1. **enzyme induction and inhibition** and 2. **genetics of the patient.** In this brick we are going to further discuss enzyme induction/inhibition, while how genetics plays a role in metabolism is discussed in "Pharmacodynamics (3 of 3) - Pharmacogenomics brick."

CYP inhibitors are drugs that reduce or inhibit enzymatic activity of the CYP protein and result in reduced biotransformation. The enzyme occurs through covalent and reversible inhibition. Drugs that are CYP inhibitors will decrease enzyme function, therefore decreasing the ability to metabolize the drug (or substrate). In other words, drugs that are substrates for the CYP inhibitor will have a *decreased rate of metabolism* and have *increased serum drug levels*. Table 1 lists some clinically **must know list !!)**

Table 1: Important CYP inhibitors

Inhibitors (+)
Cyclosporin (used as Immunosuppresant)
Cobicistat (Antiretroviral, used in HIV treatment)
Omeperazole (Proton Pump Inhibitors) (Used for Acid reflux)
Macrolides (Azithromycin, Clarithromycin) (Used as antibiotic)
Isoniazid (Used in Tuberculosis treatment)
Ciprofloxacin (Antibiotic)
Ketoconazole (Azole antibiotic - used as antifungals)
Amiodarone (Used for cardiovascular agent, antiarrhythmic)
Sulfonamides (Sulflamethoxzole/Trimethoprim)
Grapefruit juice (in large quantities)

Mnemonic

COMIC KASE

Some drugs can *induce* P450 expression by either increasing its synthesis rate or decreasing its degradation rate. These drugs are **CYP inducers.** Drugs that are CYP inducers will increase the enzyme function. In other words, drugs that are the same substrate for the CYP inducers will have an *increased rate of metabolism* and *decreased serum drug levels*. Table 2 lists some clinically significant CYP inducers. **(This is a must know list !!)**

Table 2: Important CYP inhibitors

Inducers (-)
Phenytoin (Used as Anticonvulsant)
Carbamazepine (Used as Anticonvulsant)
Alcohol
Rifampin (Used in Tuberculosis treatment)
Barbiturates such as phenobarbital (Used as Anticonvulsant)
St John's wort (OTC agent, has some antidepressant properties)
Smoking

Mnemonic

Please no CARBS

CLINICAL CORRELATION

When two drugs are metabolized by the same CYP enzyme, they can compete for the enzyme, leading to altered drug levels and potential adverse effects. For example, inhibitors of CYP3A4, like ketoconazole, can increase the plasma concentration of drugs metabolized by CYP3A4, potentially causing toxicity. Conversely, inducers of CYP enzymes, such as rifampicin, can decrease the effectiveness of drugs by increasing their metabolism. Understanding these interactions is essential for optimizing drug therapy and minimizing adverse effects.

What is the difference between a CYP450 inducers and CYP450 inhibitors?

CLINICAL CORRELATION

Pharmacogenetics is the study of inherited genetic differences that can affect individual responses to drugs. For example, CYP2D6 testing measures patient responsiveness to opioids such as codeine, hydrocodone, and oxycodone. Genotyping can determine if a patient is likely to be an ultra-rapid, normal, intermediate, or poor metabolizer of a particular drug.

Physiologic Modification

Physiologic factors, such as age, nutrition status, biologic sex, intestinal bacteria, and changes to enterohepatic circulation, can also influence CYP function and affect drug metabolism. In general, reduced blood flow leads to a decreased capacity to metabolize drugs, so if a patient's blood flow to the liver and kidneys is reduced, less drug is metabolized in and excreted from this patient, as compared with a patient whose blood flow is normal.

CLINICAL CORRELATION

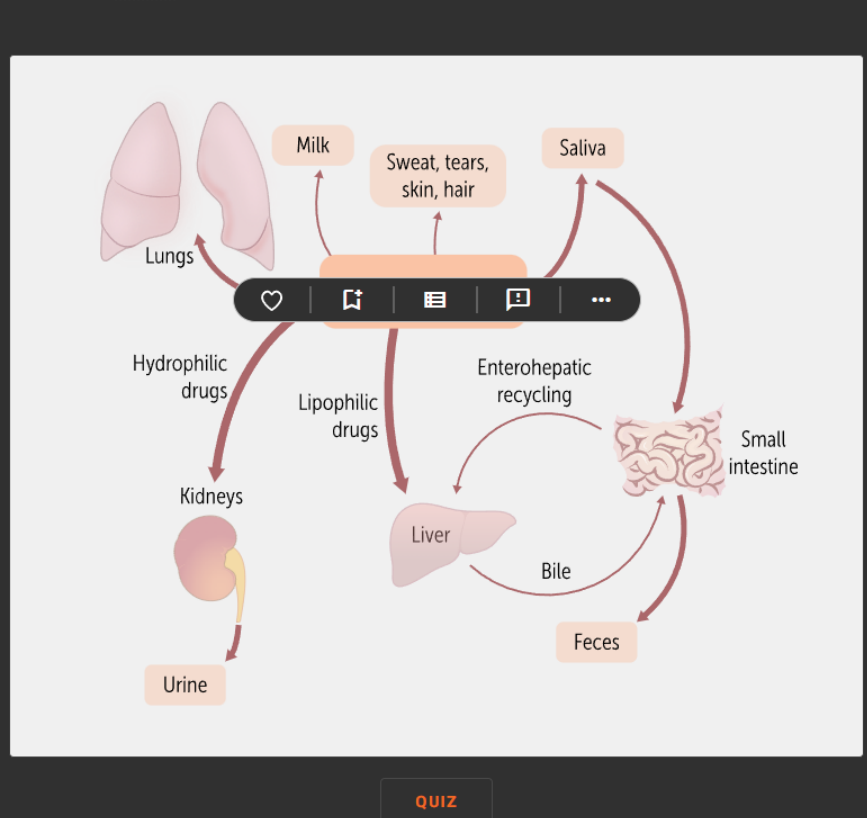
Older adults often have a reduced ability to metabolize drugs because of decreased circulation, which may require dose monitoring and adjustment.

How Are Drugs Excreted? (LO#3 and4)

We've seen that drug metabolism is the first step in removing it from the body; now let's take a deeper dive into the process of drug excretion (Figure 6). Drugs can be excreted both in unchanged form and as metabolites. Two organs are responsible for most drug excretion: the kidneys and the liver.

- The kidneys excrete water-soluble (hydrophilic) drug metabolites in the urine.
- The liver excretes water-insoluble (lipophilic) drug metabolites in the feces.

A small percentage of drugs are excreted via the lungs, skin, tears, maternal milk, and saliva.



QUIZ

Figure 6

CLINICAL REASONING

A term that is used in ADME context is drug elimination. Drug elimination is NOT the same as drug excretion. Drug elimination refers to the process of removing an administered drug from the body. It can occur through two main pathways: excretion of the drug in its unchanged form or metabolic biotransformation followed by excretion. On the other hand, excretion is the removal of the drug primarily through the kidneys. Although the kidneys are the primary organs responsible for excretion, other organs such as skin, liver, salivary glands, also can help excrete drugs.

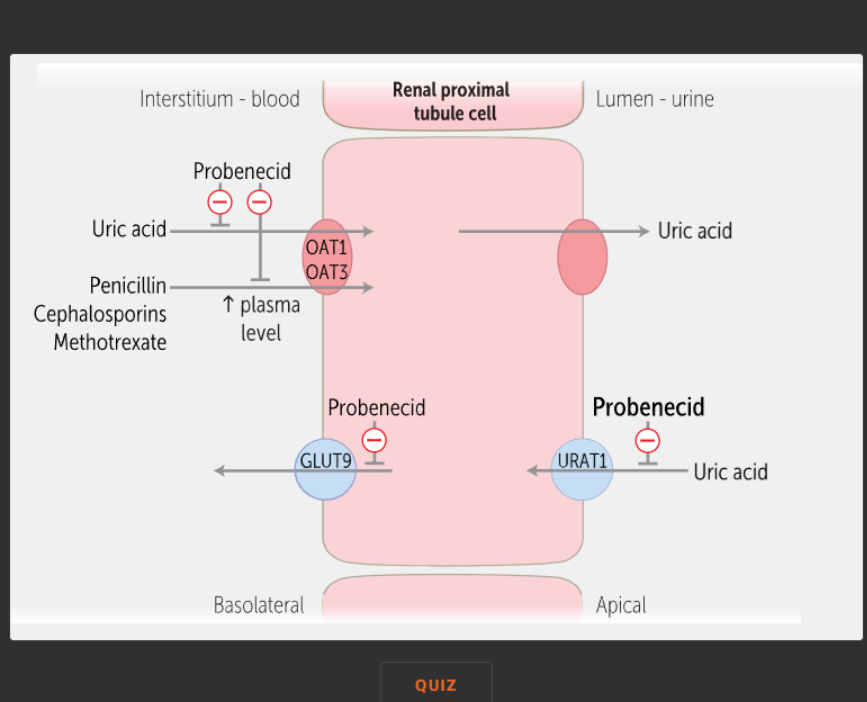
Which drugs are eliminated by the kidneys and which by the liver?

Renal Excretion

Drugs excreted via the kidneys must be unbound from large carrier proteins like albumin, since they cannot reach the urine in this form. They must also be water soluble, since urine is mostly made of water. A drug, like any chemical, reaches the urine through two main mechanisms: glomerular filtration and tubular secretion.

Some drugs can partially defeat this excretory pathway by passively diffusing back through the renal tubules from the urine into the blood.

Just like one drug can affect another drug's metabolism, it can also affect another drug's excret necid is used to decrease renal excretion of drugs such as cephalosporins, penicillin, and methotrexate. It does so by preventing secretion of these drugs into the renal tubules for elimination by binding to and inhibiting the actions of organic anion transporters (OATs) in the nephron. GLUT9 and URAT1 are uric acid transporters (Figure 7).



QUIZ

Figure 7

Credit: ©ScholarRx

CLINICAL CORRELATION

Probenecid is a uricosuric agent that inhibits tubular urate transport and increases the urine concentration of uric acid, thereby lowering its concentration in the blood and reducing gout symptoms (joint swelling and pain).

Hepatic Excretion

A few drugs are excreted by the hepatocytes into the bile, which enters the gut and is excreted in the feces. Drugs that have had a glucuronide group added in the liver enter the stool, have the glucuronide group removed by colonic bacteria, are absorbed back into the circulation by the gut, and are then re-metabolized by the liver and excreted in urine. This circulation of the drug from liver to colon back to liver is known as **enterohepatic circulation.** A few drugs can also be directly secreted into the intestine from the blood, bypassing the biliary route.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to AK, how do you explain the counseling to not drink Grapefruit juice with his new medications?

You tell AK that Grapefruit juice can alter how the medications affect the body. You explain to AK that the harmful effects of the drugs might result if Grapefruit juice if taken with the drugs. You know from the reading that Grapefruit juice is a CYP inhibitor. It can, therefore, inhibit the CYPs metabolizing simvastatin (statin) and verapamil (calcium effects. While drugs can interact with other drugs, food can also sometimes interact with drugs to potentially cause drug-related side effects.

Summary

- Drug administration, metabolism, and excretion (ADME) describe the journey of a drug through the body, with metabolism primarily occurring in the liver to make drugs inactive and water-soluble for excretion.
- Intravenous (IV) administration provides the highest bioavailability, while drug metabolism transforms drugs into compounds for easier elimination, primarily through liver enzymes.
- Hepatic biotransformation involves Phase I reactions (oxidation, reduction, hydrolysis) by CYP450 enzymes and Phase II reactions (conjugation) by hepatic transferase enzymes.
- CYP450 inducers increase enzyme activity, leading to increased drug metabolism and decreased serum drug levels, while CYP450 inhibitors decrease enzyme activity, reducing drug metabolism and increasing serum drug levels.
- Drug excretion primarily occurs through the kidneys for water-soluble drugs and the liver for lipophilic drugs, with additional minor excretion routes including lungs, skin, and saliva.
- Factors affecting drug metabolism include enzyme induction/inhibition, genetic polymorphisms, age, nutrition, sex, and physiologic modifications such as changes in blood flow.

Review Questic

1. What is the effect of CYP inhibitors on drug metabolism?

A. ● Decreased metabolism of the drug
B. ○ Increased metabolism of the drug
C. ○ Increased elimination of the drug metabolite
D. ○ Decreased excretion of the drug metabolite

Hide Explanation

The correct answer is decreased metabolism of the drug (A). CYP inhibitors reduce enzymatic activity, leading to decreased metabolism of the drug. Increased metabolism of the drug (B) is incorrect because CYP inhibitors do not enhance enzyme activity. Increased elimination of the drug metabolite (C) is also incorrect as CYP inhibitors reduce biotransformation. Decreased excretion of the drug metabolite (D) is incorrect because of decreased enzymatic metabolism.

2. Which of the following medications would most likely undergo hepatic first-pass metabolism?

A. ○ Intramuscular cefepime
B. ○ Intravenous phenobarbital
C. ● Oral estrogen
D. ○ Subcutaneous insulin
E. ○ Transdermal meperidine

Hide Explanation

The correct answer is oral estrogen (C). First-pass metabolism describes the modification of drugs either in the GI tract or after GI absorption in the first pass through the liver. It does not occur with drugs administered by non-oral routes, such as intramuscular cefepime (A), intravenous phenobarbital (B), subcutaneous insulin (D), or transdermal meperidine (E).

3. A 49-year-old male with gout controlled with probenecid was recently prescribed a cephalosporin antibiotic for pneumonia. Which of the following cephalosporin drug excretion pathways is inhibited by probenecid?

A. ○ Biliary excretion
B. ○ Conjugation with acetyl group
C. ○ Glomerular filtration
D. ○ Passive diffusion into tubules
E. ● Renal secretion

Hide Explanation

The correct answer is renal secretion (E). Probenecid blocks the OATs, which are responsible for active secretion of drugs into renal tubules for elimination. Thus, probenecid would decrease drug excretion, resulting in increased drug levels in the blood. Biliary excretion (A) can help eliminate lipid-soluble drugs and is not affected by probenecid. Conjugation with an acetyl group (B) is a step of drug metabolism, not drug excretion. Glomerular filtration (C) is a form of renal drug excretion of unbound drugs but is not affected by probenecid. Passive diffusion into tubules (D) is another way that drugs are renally excreted and is not affected by probenecid.

4. Which organ primarily metabolizes drugs?

A. ○ Kidneys
B. ○ Skin
C. ○ Lungs
D. ● Liver

Hide Explanation

The correct answer i ration of drug-metabolizing enzymes. The Kidneys (A), Skin (B), and Lungs (C) are primarily responsible for drug excretion and not for drug metabolism.

5. Which of the following is a CYP inhibitor?

A. ○ Carbamazepine
B. ● Ciprofloxacin
C. ○ Phenobarbital
D. ○ Rifampin

Hide Explanation

The correct answer is ciprofloxacin (B), a CYP450 inhibitor that would reduce the metabolic effect of the enzyme, Carbamazepine (A), phenobarbital (C), and rifampin (D) are all inducers of CYP450.

6. Which organ excretes lipophilic (water-insoluble) drug metabolites?

A. ○ Kidneys
B. ● Liver
C. ○ Lungs
D. ○ Skin

Hide Explanation

The correct answer is the liver (B). The liver excretes water-insoluble (lipophilic) drug metabolites in the feces. The kidneys (A) excrete water-soluble (hydrophilic) drug metabolites in the urine. The lungs (C), skin (D), and spleen (E) are not primary organs for drug excretion.

7. What is the effect of CYP inducers on serum drug levels?

A. ○ Increased serum drug levels
B. ○ No change in serum drug levels
C. ○ Increased half-life
D. ● Decreased serum drug levels
E. ○ Decreased drug metabolism

Hide Explanation

The correct answer is decreased serum drug levels (D). CYP inducers increase the enzyme function, lead nd decreased serum drug levels. Increased e it is the effect of CYP inhibitors. No change in serum drug levels (B) is incorrect as CYP inducers do affect drug levels. Increased drug half-life (D) is incorrect because CYP inducers decrease the drug half-life. Decreased drug metabolism (E) is incorrect because CYP inducers increase drug metabolism.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01	☑	~~Distinguish between Phase I and Phase II reactions in hepatic drug metabolism.~~
02	☑	~~Determine which drugs act as substrates and identify the key enzymes implicated in their hepatic biotransformation.~~
03	☑	~~Explain the role of various organs in drug excretion.~~
04	☑	~~Identify relevant pharmacokinetic principles and apply those to solve clinical problems and predict outcomes.~~

Objective Confidence: 4/4

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